Continuation of the C^*-Algebra Notes
Finite-Dimensional Corners, Compact Operators, and the Calkin Algebra

These notes continue the rank-one and corner calculations developed in the previous session. The handwritten order has been preserved, while a few short explanations have been added to make the transitions explicit. Such additions are marked in the text or in footnotes.

Throughout, let H be an infinite-dimensional separable complex Hilbert space with fixed orthonormal basis

$$(e_1, e_2, \ldots).$$

We keep the convention from the preceding notes that the inner product is linear in the first variable, and

$$(e_i \otimes e_j)(h) = \langle h, e_j \rangle e_i.$$

The assumption that the basis is indexed by $\mathbb{N}$ is exactly the separability assumption used implicitly in the handwritten notes.[1]

1 The rank-two corner revisited

Let

$$p = e_1 \otimes e_1 + e_2 \otimes e_2.$$

If $S \in \mathcal{B}(H)$, write

$$S(e_j) = \sum_{k=1}^{\infty} \alpha_{kj} e_k, \qquad j = 1, 2.$$

Using the matrix-unit relation

$$(e_i \otimes e_j)(e_r \otimes e_s) = \delta_{jr}\, e_i \otimes e_s,$$

one obtains

$$pSp = \alpha_{11} e_1 \otimes e_1 + \alpha_{12} e_1 \otimes e_2 + \alpha_{21} e_2 \otimes e_1 + \alpha_{22} e_2 \otimes e_2.$$

Therefore the map

$$pSp \longmapsto \begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{pmatrix}$$

identifies the corner $p\mathcal{B}(H)p$ with $M_2(\mathbb{C})$ as a C^*-algebra.[2]

[1] For a nonseparable Hilbert space one usually replaces the sequence (p_n) below by a directed family of finite-dimensional projections. The definition of $\mathcal{K}(H)$ remains the same, but some statements about the ideal structure of $\mathcal{B}(H)$ require additional care.

[2] In the handwritten notes the four coefficients are also displayed as a point of $\mathbb{C}^4$. That is correct as a *vector-space* identification. The algebra product, however, is the matrix product, so the natural algebraic identification is $p\mathcal{B}(H)p \cong M_2(\mathbb{C})$, not coordinatewise multiplication on $\mathbb{C}^4$.

2 The finite-dimensional projections p_n

For $n \in \mathbb{N}$, define

$$p_n := \sum_{i=1}^{n} e_i \otimes e_i.$$

Then p_n is the orthogonal projection onto

$$H_n := \operatorname{span}\{e_1, \ldots, e_n\}.$$

Proposition. For every $n \geq 1$,

$$p_n \mathcal{B}(H) p_n \cong M_n(\mathbb{C}).$$

Indeed, if

$$T(e_j) = \sum_{k=1}^{\infty} \alpha_{kj} e_k,$$

then

$$p_n T p_n = \sum_{i,j=1}^{n} \alpha_{ij}\, e_i \otimes e_j.$$

Hence the $*$-isomorphism is given by

$$\Phi_n : p_n \mathcal{B}(H) p_n \longrightarrow M_n(\mathbb{C}), \qquad \Phi_n(p_n T p_n) = [\alpha_{ij}]_{i,j=1}^{n}.$$

The unit of the corner is p_n, and under Φ_n it corresponds to I_n. This is the standard realization of $M_n(\mathbb{C})$ as a finite-dimensional corner of $\mathcal{B}(H)$.[3]

2.1 Compatibility of the corners

The handwritten notes record the inclusion

$$p_n \mathcal{B}(H) p_n \subseteq p_{n+1} \mathcal{B}(H) p_{n+1}.$$

To see it, first note that

$$\begin{aligned} p_n p_{n+1} &= \left(\sum_{i=1}^{n} e_i \otimes e_i \right) \left(\sum_{j=1}^{n+1} e_j \otimes e_j \right) \\ &= \sum_{i=1}^{n} e_i \otimes e_i = p_n, \end{aligned}$$

and similarly $p_{n+1} p_n = p_n$. Thus, for $T \in \mathcal{B}(H)$,

$$p_{n+1}(p_n T p_n) p_{n+1} = p_n T p_n.$$

Therefore $p_n T p_n$ belongs to the $(n+1)$-st corner.

Under the identifications $p_n \mathcal{B}(H) p_n \cong M_n(\mathbb{C})$, this is exactly the usual block embedding

$$M_n(\mathbb{C}) \hookrightarrow M_{n+1}(\mathbb{C}), \qquad x \longmapsto \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}.$$

Consequently,

$$\bigcup_{n=1}^{\infty} M_n(\mathbb{C}) \cong \bigcup_{n=1}^{\infty} p_n \mathcal{B}(H) p_n \subseteq \mathcal{B}(H).$$

This union is an increasing $*$-subalgebra of $\mathcal{B}(H)$ consisting of operators whose matrix, relative to the chosen basis, has support in a finite upper-left block.[4]

[3] Compare Davidson, [?, §I.10], and Erdman, [?, §7.3–7.4].

[4] For a fixed basis this algebraic union need not equal all of $\mathcal{F}(H)$: a rank-one operator $\xi \otimes \eta$ may involve

3 Finite-rank operators

Let $\mathcal{F}(H)$ denote the linear space of finite-rank operators in $\mathcal{B}(H)$.

Proposition. If $T \in \mathcal{F}(H)$, then T is a finite sum of rank-one operators. More precisely, if

$$\{e_1, \ldots, e_n\}$$

is an orthonormal basis for $\operatorname{ran} T$, then

$$\boxed{T = \sum_{i=1}^{n} e_i \otimes T^*(e_i).}$$

Proof. For $h \in H$, since $Th \in \operatorname{ran} T$,

$$\begin{aligned} Th &= \sum_{i=1}^{n} \langle Th, e_i \rangle e_i \\ &= \sum_{i=1}^{n} \langle h, T^* e_i \rangle e_i \\ &= \left(\sum_{i=1}^{n} e_i \otimes T^*(e_i) \right)(h). \end{aligned}$$

Therefore

$$T = \sum_{i=1}^{n} e_i \otimes T^*(e_i).$$

□

This proves, in particular, that the finite-rank operators are exactly the finite linear combinations of rank-one operators.[5]

4 The algebra of compact operators

Definition. The C^*-algebra of compact operators on H is

$$\boxed{\mathcal{K}(H) := \overline{\mathcal{F}(H)}^{\|\cdot\|}.}$$

That is, $\mathcal{K}(H)$ is the operator-norm closure of the finite-rank operators.

On a Hilbert space this norm-closure definition agrees with the usual topological definition: $T \in \mathcal{B}(H)$ is compact if it maps the closed unit ball into a relatively compact subset of H.[6]

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vectors ξ, η with infinitely many nonzero basis coefficients. What is true, and what is used below, is that the norm closure of this increasing union is $\mathcal{K}(H)$.

[5] A standard reference is Conway, [**?**, Chapter II, §4]; see also the explicit rank-one discussion in Erdman, [**?**, §7.3].

[6] This equivalence is one of the standard basic results on compact operators; see Conway, [**?**, Chapter II, Theorem 4.4], or MIT 18.102, Lecture 18, [**?**].

4.1 Compact operators from the matrix corners

The finite-dimensional corners form a norm-dense subalgebra of $\mathcal{K}(H)$:

$$\boxed{\mathcal{K}(H) = \overline{\bigcup_{n=1}^{\infty} p_n \mathcal{B}(H) p_n}^{\|\cdot\|} \cong \overline{\bigcup_{n=1}^{\infty} M_n(\mathbb{C})}^{\|\cdot\|}.}$$

Here each $M_n(\mathbb{C})$ is embedded in $M_{n+1}(\mathbb{C})$ by

$$x \mapsto \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}.$$

The closure is essential: the algebraic union contains only finite matrices, whereas $\mathcal{K}(H)$ also contains norm limits of such matrices.[7]

A useful approximation statement is

$$T \in \mathcal{K}(H) \quad \Longrightarrow \quad \|T - p_n T p_n\| \longrightarrow 0.$$

For a compact operator this follows from the strong convergence $p_n \to I$ together with uniform convergence on compact subsets. This supplies the link between the handwritten increasing matrix corners and the full compact-operator algebra.

4.2 $*$-closedness

Suppose $T_m \in \mathcal{F}(H)$ and

$$T_m \longrightarrow T \qquad \text{in operator norm.}$$

Since finite rank is preserved by taking adjoints,

$$T_m^* \in \mathcal{F}(H).$$

Moreover,

$$\|T_m^* - T^*\| = \|T_m - T\| \longrightarrow 0.$$

Therefore

$$T \in \mathcal{K}(H) \quad \Longrightarrow \quad T^* \in \mathcal{K}(H).$$

Thus $\mathcal{K}(H)$ is closed under the involution.

The inclusions

$$M_n(\mathbb{C}) \subseteq \mathcal{K}(H) \qquad (n \geq 1)$$

show that $\mathcal{K}(H)$ is infinite-dimensional whenever H is infinite-dimensional, since

$$\dim_{\mathbb{C}} M_n(\mathbb{C}) = n^2 \longrightarrow \infty.$$

5 $\mathcal{K}(H)$ as an ideal in $\mathcal{B}(H)$

Proposition. If $T \in \mathcal{K}(H)$ and $S \in \mathcal{B}(H)$, then

$$TS \in \mathcal{K}(H) \qquad \text{and} \qquad ST \in \mathcal{K}(H).$$

Consequently, $\mathcal{K}(H)$ is a closed two-sided $*$-ideal of $\mathcal{B}(H)$.

[7] This viewpoint is the first model of an inductive-limit construction and is closely related to the later theory of AF C^*-algebras. Davidson's text develops this point of view systematically; see [**?**, Chapters I and III].

Proof. Choose finite-rank operators T_m such that

$$\|T_m - T\| \longrightarrow 0.$$

Then $T_m S$ and $S T_m$ are finite rank, while

$$\|T_m S - TS\| \leq \|T_m - T\| \, \|S\| \longrightarrow 0,$$
$$\|S T_m - ST\| \leq \|S\| \, \|T_m - T\| \longrightarrow 0.$$

Hence $TS, ST \in \mathcal{K}(H)$. □

The handwritten sentence that $\mathcal{K}(H)$ is "the unique C^*-ideal of $\mathcal{B}(H)$" should be read more precisely as follows.[8]

Proposition. If H is separable and infinite-dimensional, then $\mathcal{K}(H)$ is the unique nonzero proper norm-closed two-sided ideal of $\mathcal{B}(H)$.

This separability hypothesis matters: for nonseparable Hilbert spaces, $\mathcal{B}(H)$ has additional nontrivial closed ideals.[9]

6 The Calkin algebra

Definition. The *Calkin algebra* is the quotient C^*-algebra

$$\boxed{\mathcal{Q}(H) := \mathcal{B}(H)/\mathcal{K}(H).}$$

If

$$\pi : \mathcal{B}(H) \longrightarrow \mathcal{Q}(H), \qquad \pi(T) = T + \mathcal{K}(H),$$

denotes the quotient map, then the inclusion

$$\iota : \mathcal{K}(H) \hookrightarrow \mathcal{B}(H)$$

and π give the short exact sequence

$$\boxed{0 \longrightarrow \mathcal{K}(H) \xrightarrow{\iota} \mathcal{B}(H) \xrightarrow{\pi} \mathcal{Q}(H) \longrightarrow 0.}$$

Thus the Calkin algebra identifies two bounded operators whenever they differ by a compact operator. In this sense it records operator-theoretic information "modulo compact perturbations."[10]

[8] The ideals $\{0\}$ and $\mathcal{B}(H)$ are always present, so "unique" here means unique *nonzero proper norm-closed two-sided ideal*.

[9] For a precise statement, see Blackadar, [**?**, §I.8.7]. In particular, he explicitly distinguishes the separable and nonseparable cases.

[10] For later study: Atkinson's theorem says that $T \in \mathcal{B}(H)$ is Fredholm if and only if its coset $\pi(T)$ is invertible in the Calkin algebra. This theorem was not written in the photographs, but it explains why the quotient $\mathcal{B}(H)/\mathcal{K}(H)$ becomes central in Fredholm theory; see Murphy, [**?**, §1.4], or Blackadar, [**?**, Chapter I].

Reading guide for the photographs

The mathematical chain represented in the photographs is

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\boxed{
\begin{aligned}
p_n\mathcal{B}(H)p_n \cong M_n(\mathbb{C}), \qquad & M_n(\mathbb{C}) \hookrightarrow M_{n+1}(\mathbb{C}),\\
\bigcup_{n\geq 1} p_n\mathcal{B}(H)p_n \text{ is norm dense in } & \mathcal{K}(H),\\
\mathcal{F}(H) = \operatorname{span}\{\text{rank-one operators}\}, \qquad & \mathcal{K}(H) = \overline{\mathcal{F}(H)}^{\|\cdot\|},\\
\mathcal{K}(H) \triangleleft \mathcal{B}(H), \qquad & \mathcal{Q}(H) = \mathcal{B}(H)/\mathcal{K}(H),\\
0 \longrightarrow \mathcal{K}(H) \longrightarrow \mathcal{B}(H) \longrightarrow \mathcal{Q}(H) \longrightarrow 0.&
\end{aligned}
}
$$

The sequence should be read as a passage from finite matrices to finite-rank operators, then to their norm closure, and finally to the quotient that forgets compact perturbations.

References and where to look

References

[1] J. B. Conway, *A Course in Functional Analysis*, 2nd ed., Graduate Texts in Mathematics 96, Springer, 1990. See Chapter II, especially §4 "Compact Operators."

[2] G. J. Murphy, *C^*-Algebras and Operator Theory*, Academic Press, 1990. See §1.4 for compact and Fredholm operators and Chapter 2 for Hilbert-space operator C^*-algebras.

[3] K. R. Davidson, *C^*-Algebras by Example*, Fields Institute Monographs 6, American Mathematical Society, 1996. See §I.10, "C^*-Algebras of Compact Operators," and Chapter III for AF algebras and inductive-limit ideas.

[4] B. Blackadar, *Operator Algebras: Theory of C^*-Algebras and von Neumann Algebras*, Encyclopaedia of Mathematical Sciences 122, Springer, 2006. See §I.8 on compact operators and §I.8.7 on ideals of $\mathcal{B}(H)$.

[5] J. M. Erdman, *Lecture Notes on Operator Algebras*, Portland State University, 2011. See §7.3 "Finite Rank Operators" and §7.4 "Compact Operators." Available online from Portland State University.

[6] R. B. Melrose, *18.102 Introduction to Functional Analysis — Lecture 18: Compact Operators*, MIT OpenCourseWare, Spring 2009. The lecture proves that compact operators on a separable Hilbert space are exactly the norm limits of finite-rank operators.